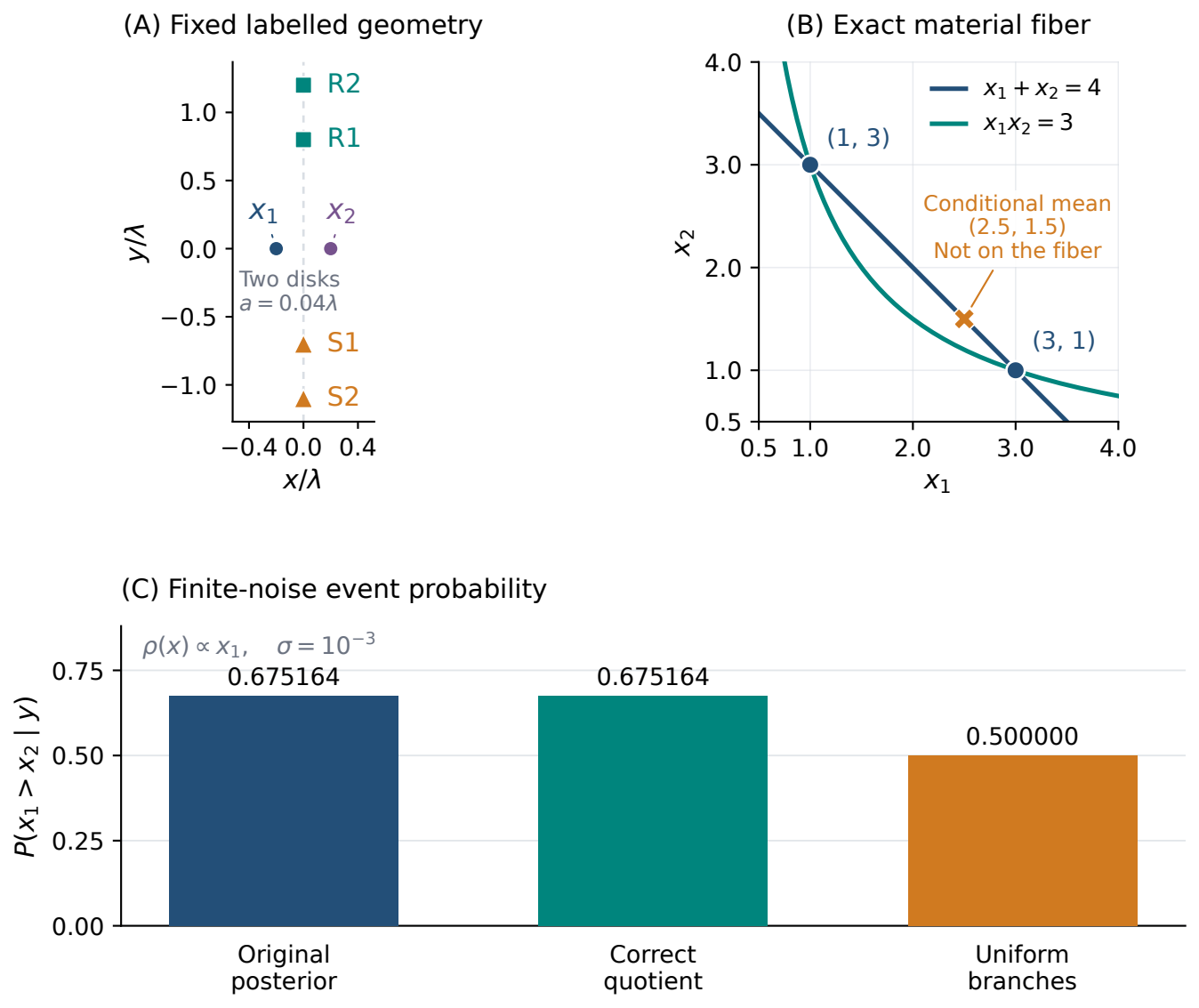
(A) Fixed labelled geometry
R2
R1
x_1
x_2
Two disks
$a = 0.04\lambda$
S1
S2
y/λ
x/λ
1.0
0.5
0.0
−0.5
−1.0
−0.4
0.0
0.4
(B) Exact material fiber
$x_1 + x_2 = 4$
$x_1 x_2 = 3$
(1, 3)
Conditional mean
(2.5, 1.5)
Not on the fiber
(3, 1)
x_2
x_1
4.0
3.0
2.0
1.0
0.5
0.5
1.0
2.0
3.0
4.0
(C) Finite-noise event probability
$\rho(x) \propto x_1, \quad \sigma = 10^{-3}$
$P(x_1 > x_2 \mid y)$
0.675164
0.675164
0.500000
0.75
0.50
0.25
0.00
Original posterior
Correct quotient
Uniform branches